

Previous Test and Exam Problems

Otto/Diesel Cycle

Problem 1

An ideal Otto cycle has a compression ratio of 8.5. At the beginning of the compression process, air is at 100 kPa and 20°C. The heat added during the heat addition process is 800 kJ/kg.

- (a) Sketch a P - v and a T - s diagram for a typical Otto cycle and list the four processes that make up the ideal Otto cycle. (4 marks)

Using the data above and assuming constant specific heat values at room temperature (approximate analysis), determine:

- (b) The pressure and temperature at the end of the heat addition process. (6 marks)
- (c) The heat rejected during the heat rejection process per kg of air. (3 marks)
- (d) The net work output per kg of air, using your calculated heat values. (2 marks)
- (e) The thermal efficiency of the cycle, using your calculated values of heat and work. (*confirm* your answer using $\eta_{th,Otto}$ from formula sheet). (2 marks)
- (f) The mean effective pressure for the cycle. (3 marks)

Cold air standard assumptions:

$$k = 1.4 \quad c_p = 1.005 \text{ kJ/kgK} \quad c_v = 0.718 \text{ kJ/kgK}$$

Problem 2

An air-standard diesel cycle has a compression ratio of 20 and a cut-off ratio of 2.2. The air at the beginning of the compression process is at 98 kPa and 25°C (Figure 4). Using constant specific heats (approximate analysis), determine:

- (a) The maximum cycle temperature. (4 marks)
- (b) The heat supplied during heat addition process. (2 marks)
- (c) The temperature at the end of the expansion process. (2 marks)
- (d) The net-work output per kg of air. (4 marks)
- (e) The thermal efficiency of the cycle (*confirm* your answer using $\eta_{th,Diesel}$ from formula sheet). (4 marks)
- (f) The mean effective pressure (MEP) for the cycle. (4 marks)

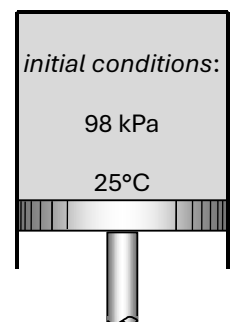


Figure 4: Diesel cycle

Problem 3

The compression ratio of an air-standard Diesel cycle is 19.5. The state of the air at the beginning of the compression process is 100 kPa and 15°C. If the maximum temperature of the process is not to exceed 1900 K, and using constant specific heats at room temperature (approximate analysis), determine:

- (g) The cut-off ratio (4 marks)
- (h) The heat supplied during heat addition process. (2 marks)
- (i) The temperature at the end of the expansion process. (2 marks)
- (j) The net-work output per kg of air for one complete cycle. (4 marks)
- (k) The thermal efficiency of the cycle (confirm your answer using $\eta_{th,Diesel}$ from formula sheet). (4 marks)
- (l) The mean effective pressure (MEP) for the cycle. (4 marks)

Brayton Cycle**Problem 4**

A turbojet aircraft flies with a velocity of 900 km/h at an altitude where the air temperature and pressure are -35°C and 40 kPa. Air leaves the diffuser at 50 kPa with a velocity of 15 m/s, and combustion gases enter the turbine at 450 kPa and 950°C. The turbine produces 500 kW of power, all of which is used to drive the compressor. The isentropic efficiency of the compressor, turbine, and nozzle is 83 percent. Assume constant specific heats with $c_p = 1005$ kJ/kg and $k = 1.4$.

- a) Sketch the cycle in a $T-s$ diagram. (5 marks)
- b) Determine the pressure of the combustion gases at the turbine exit. (7 marks)
- c) What is the mass flow rate of air through the compressor? (2 marks)
- d) Assuming negligible velocity of the combustion gases at turbine exit, determine the velocity of the gases at the nozzle exit. (6 marks)

Problem 5

A regenerator with an effectiveness of 85% is added to an open Brayton cycle (Figure 1). Air enters the compressor of this cycle at 90 kPa and 10°C; the pressure ratio is 8; and the maximum cycle temperature is 815°C. Both the compressor and the turbine have an isentropic efficiency of 90%.

Assume constant properties for air with $c_v = 0.718$ kJ/kg·K, $c_p = 1.005$ kJ/kg·K, $k = 1.4$, $R = 0.287$ kJ/kg·K.

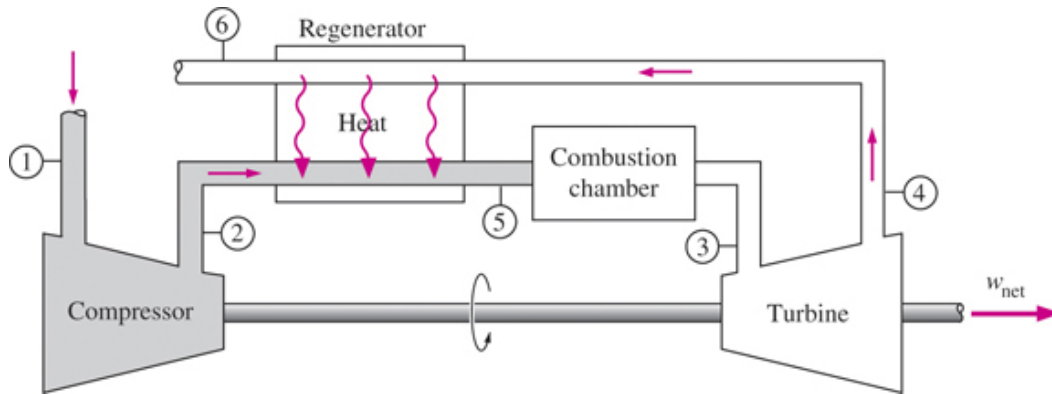


Figure 1: Open Brayton cycle with regenerator

- Sketch and label the cycle in a T-s diagram. (4 marks)
- What is the specific net work of this gas turbine? (10 marks)
- What is the thermal efficiency of this system? (4 marks)
- What would the thermal efficiency of this cycle be without a regenerator? (2 marks)

Problem 6

A gas turbine (Brayton) cycle with a regenerator, shown in Figure 1, is designed to produce a net power output of 150 kW. Air enters the compressor at 100 kPa and 20°C. The compressor pressure ratio is 8; the maximum cycle temperature is 800°C; compressor isentropic efficiency is 87%, and the turbine isentropic efficiency is 93%. With respect to the regenerator, the 'cold' air stream (State 5) leaves the regenerator 10°C lower than that of the hot stream at the inlet (State 4).

Assume constant properties for air at 300 K with $c_v = 0.718$ kJ/kg·K, $c_p = 1.005$ kJ/kg·K, $R = 0.287$ kJ/kg·K, and $k = 1.4$.

- Sketch the T-s diagram for the cycle. Show the numbers in Figure 1 on your sketch. (2 marks)
- Determine the mass flow rate of air through the system. (5 marks)
- Determine the thermal efficiency of the system. (6 marks)
- Determine the effectiveness of the regenerator. (2 marks)

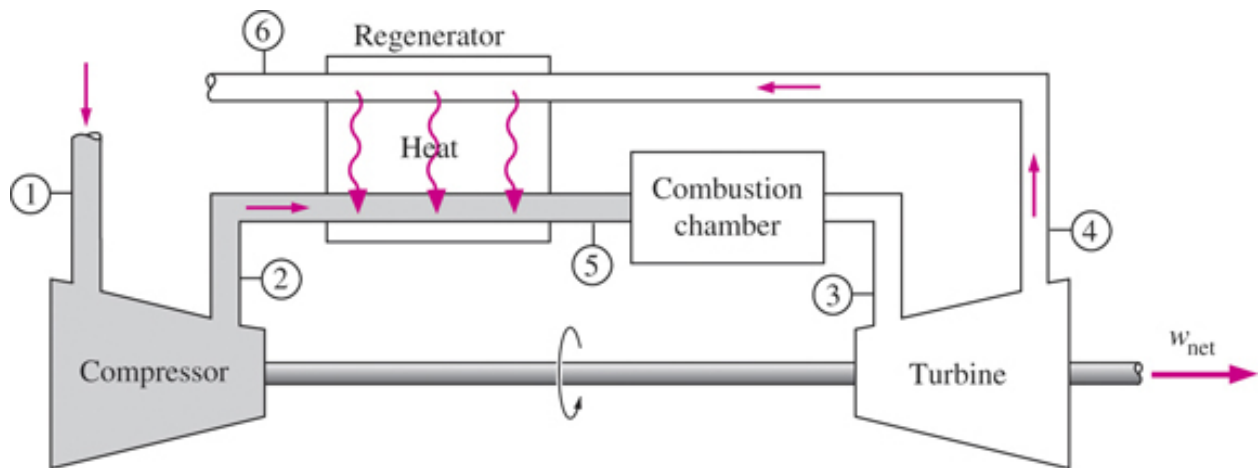


Figure 1

Rankine Cycle

Problem 7

A steam power plant operates on a Rankine cycle with an open feedwater heater (Figure 2). Steam enters the turbine at 15 MPa and 600°C at a rate of 84 kg/s and is condensed in the condenser at a pressure of 10 kPa. Some steam leaves the turbine at a pressure of 1.2 MPa and enters the open feedwater heater. Assume that the turbine and pump are isentropic devices.

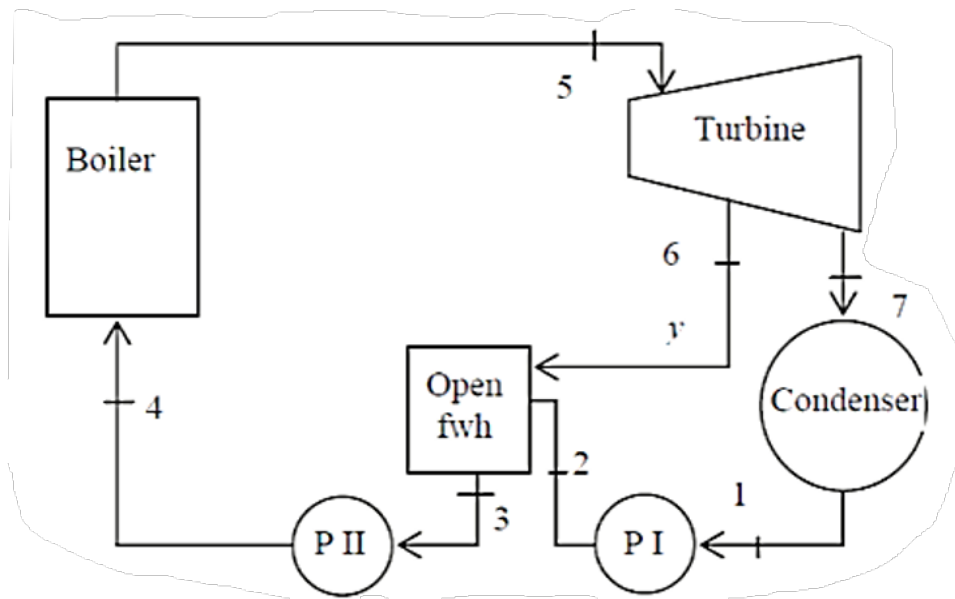
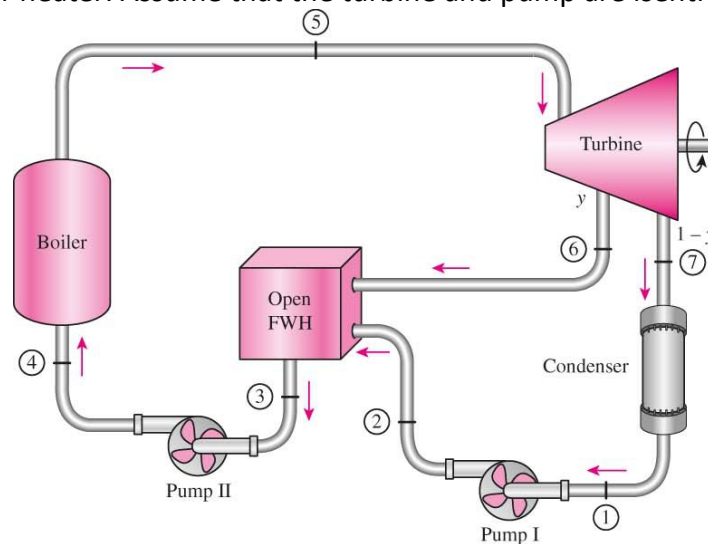


Figure 2: Regenerative Rankine cycle with Open Feedwater Heater

- Sketch and label a temperature-entropy (T - s) diagram showing this cycle (4 marks)
- Determine the fraction of steam extracted from the turbine for feedwater heating. (14 marks)
- Determine the power produced by the plant. (9 marks)

Problem 8

A steam power plant operates on a Rankine cycle with an open feedwater heater. Steam enters the turbine at 15 MPa and 600°C at a rate of 84 kg/s and is condensed in the condenser at a pressure of 10 kPa. Some steam leaves the turbine at a pressure of 1.2 MPa and enters the open feedwater heater. Assume that the turbine and pump are isentropic devices.



- Sketch and label a temperature-entropy (T - s) diagram showing this cycle. (4 marks)
- Determine the fraction of steam extracted from the turbine for feedwater heating. (14 marks)
- Determine the power produced by the plant. (7 marks)

Problem 9

Consider a steam power plant that operates on the ideal regenerative Rankine cycle with a closed feedwater heater as shown in Figure 1. The plant maintains the turbine inlet at 8 MPa and 450°C and operates the condenser at 20 kPa. A quarter ($y = 0.25$) of the steam is extracted at 2000 kPa to serve the closed feedwater heater, which fully condenses before it is pumped to boiler pressure. It then mixes with the stream coming from the condenser.

- a) Sketch the cycle in a T - s diagram showing the saturation lines. (5 marks)

(Make a reasonable assumption on the states at turbine exits. There is no need to correct them afterwards!)

- b) Calculate the thermal efficiency of the cycle. (23 marks)

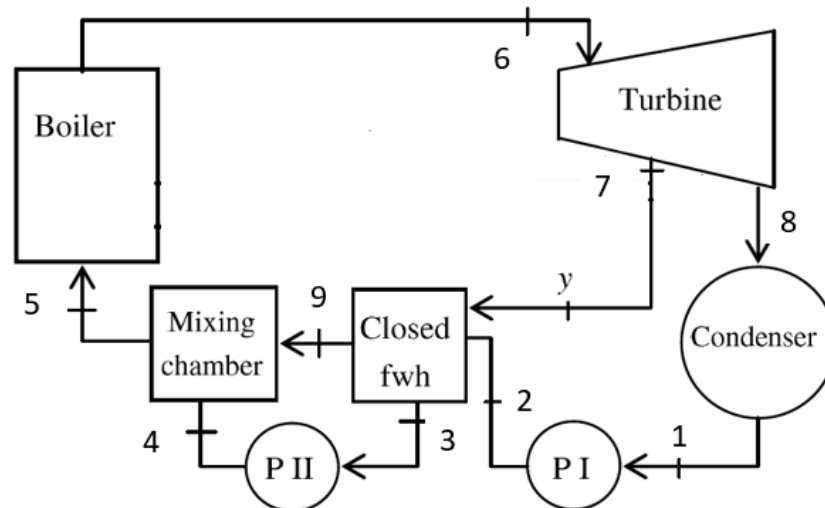


Figure 1: Regenerative Rankine cycle with a closed feedwater heater

Refrigeration

Problem 10

A commercial refrigerator with R-134a as the working fluid is used to keep the refrigerated space at -25°C by rejecting its waste heat to cooling water that enters the condenser at 20°C at a rate of 0.25 kg/s and leaves at 25°C . The refrigerant enters the condenser at 1.2 MPa and 70°C and leaves at 42°C . The inlet state of the compressor is 60 kPa and -20°C and the compressor is estimated to gain a net heat of 450 W from the surroundings.

- a) Determine the quality of the refrigerant at the evaporator inlet. (6 marks)
- b) Calculate the refrigeration load. (10 marks)
- c) What is the COP of the refrigerator? (2 marks)

Problem 11

Part A:

Briefly explain the main factors influencing the choice of evaporator pressure in a vapour-compression refrigeration system.

(2 marks)

Part B:

Consider a 250 kW refrigeration system which operates on a simple vapour-compression refrigeration cycle using Refrigerant-134a. The refrigerant enters the compressor as a saturated vapour at 140 kPa and is compressed to 800 kPa. The isentropic efficiency of the compressor is 85%. You may assume the refrigerant is a saturated liquid at the entrance to the throttle process.

- a) Sketch and label a temperature-entropy (T - s) diagram showing the cycle.

(2 marks)

- b) Determine the quality of the vapour at the end of the throttling process.

(5 marks)

- c) Determine the power input to the compressor.

(9 marks)

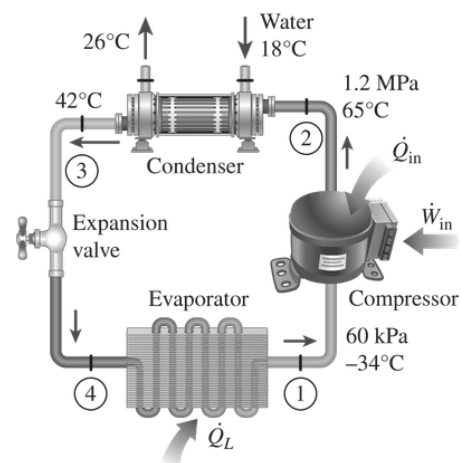
- d) Determine the coefficient of performance of the system.

(2 marks)

Problem 12

A commercial refrigerator operating on a simple vapour-compression cycle using refrigerant R-134a is used to maintain a refrigeration space at -30°C . The refrigerant enters the condenser at 1.2 MPa and 65°C and leaves at 42°C . The inlet of the compressor is at 60 kPa and -34°C , and the compressor gains about 450 W from the surroundings. Some of these data are included in the figure alongside.

The condenser rejects heat to cooling water which enters the condenser at rate of 0.25 kg/s and 18°C and leaves at 26°C . It may be assumed that the condenser acts a perfect heat exchanger with an effectiveness of 100%.



Also, you may ignore pressure losses in the condenser and evaporator and can assume that their pressures are 1.2 MPa and 60 kPa respectively.

- a) Sketch a temperature-entropy (T - s) diagram showing this cycle and clearly label positions 1, 2, 3 and 4.

(3 marks)

- b) Determine the refrigeration capacity (W) this system, the power input to the compressor and the coefficient of performance of the system. (14 marks)
- c) In an ideal vapour compression cycle, the condition at the entrance to the compressor (position 1) is a saturated vapour, while in real commercial refrigeration cycles the condition at the entrance of the compressor (position 1) should always be a superheated vapour. Briefly explain why this is necessary. (3 marks)